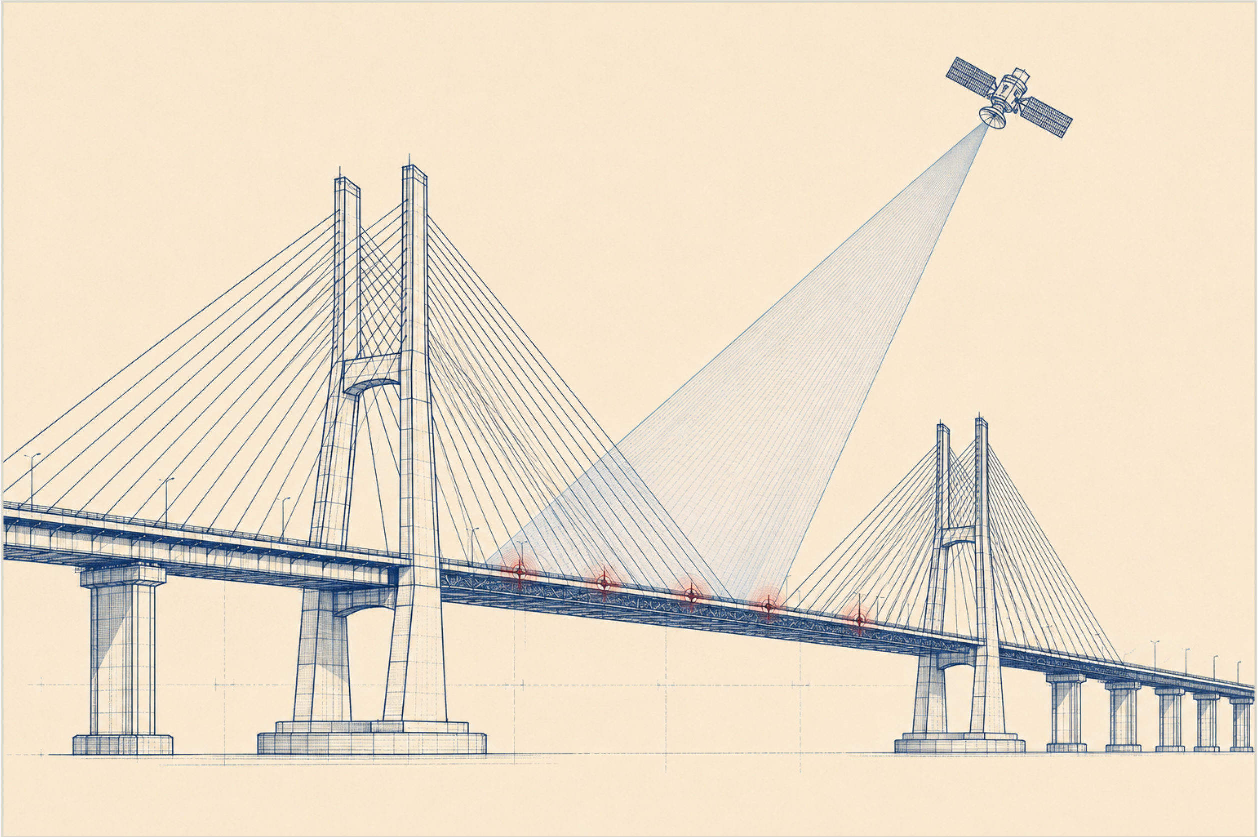
DISSENT

A machine second opinion that predicts infrastructure failure by catching the moment the physical evidence stops agreeing with the official record.



TEAM NEXUS NETWORK

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)

> **Infrastructure does not fail silently. It fails contradicted.**
> **DISSENT is the machine that files the dissent.**

PROBLEM	CORE APPROACH	PRIMARY USERS	OUTPUT
Predicting Infrastructure Failure From Weak Signals	Dual-witness disagreement scoring: record vs physics	Bridge-owning agencies: MoRTH / NHAI, state DOTs	Ranked dissent docket with dated inspection obligations

01 Problem Statement

ASSIGNED PROBLEM : ROUND 2 : VERBATIM

Develop a system that identifies infrastructure quietly approaching failure by combining traffic patterns, weather, maintenance history, satellite imagery, vibration measurements, construction records, and age.

The phrase "quietly approaching failure" is the whole difficulty. Catastrophic failures are vanishingly rare (a handful of significant bridge collapses per year against hundreds of thousands of standing structures), so a naive classifier has almost nothing to learn from. Worse, the forensic record of famous collapses shows that the warning almost never arrives as one loud alarm. It arrives as several weak signals, each below any action threshold, each held by a different organisation, none of them anyone's explicit job to combine.

623,218
US BRIDGES IN THE NATIONAL INVENTORY

6.8%
RATED POOR (ASCE 2025: GRADE C)

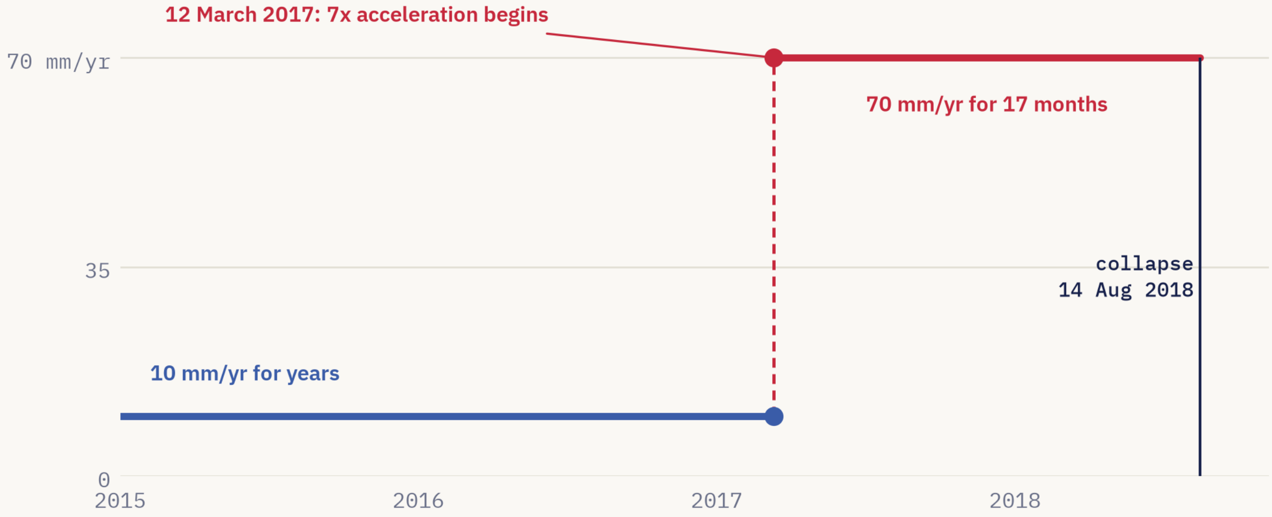
163M
DAILY CROSSINGS ON POOR BRIDGES (ARTBA)

$373B
TEN-YEAR INVESTMENT GAP (ASCE 2025)

US bridge stock by condition rating, ASCE 2025 Report Card. The 6.8% poor slice is 41,600 structures that cannot all be re-inspected tomorrow.

That weak-signal pattern is remarkably consistent across the best documented failures of the last two decades:

FAILURE	WEAK SIGNAL THAT ALREADY EXISTED	WHY NOTHING HAPPENED
Morandi Bridge Genoa 2018, 43 dead	Radar analysis published after the collapse (Milillo et al. 2019): a satellite scatterer on the deck beside the failed pier accelerated from about 10 to 70 mm/yr from March 2017, roughly 17 months before collapse; a cable strength-loss assessment already existed	Displacement data, structural assessments and retrofit paperwork sat in separate silos; the retrofit was scheduled, not urgent
I-35W Minneapolis 2007, 13 dead	Inspection photographs from 1999 and 2003 already showed visibly bowed gusset plates	The record itself said "deficient" for years; no mechanism forced escalation while load was added
FIU pedestrian bridge Florida 2018, 6 dead	Rapidly growing structural cracks, photographed and discussed by the engineer of record	A meeting held the morning of the collapse concluded the cracks were not a safety concern
Champlain Towers South Surfside 2021, 98 dead	A satellite study published in 2020 (Fiaschi and Wdowinski) had identified the site as subsiding by about 2 mm/yr in the 1990s; a 2018 engineering report documented major structural damage	The subsidence study, the damage report and the deferred repair decisions were never combined
Morbi footbridge Gujarat 2022, 135 dead	The bridge reopened four days after a renovation that never replaced or load-tested the original main cables; the inquiry later found 22 of the 49 wires in the failed cable were already corroded	No independent check existed that the renovation record matched the physical state



The Morandi precursor as published: line-of-sight velocity of the deck scatterer beside the failed pier (drawn from Milillo et al. 2019; the finding is contested by Lanari et al. 2020, and our validation replays both analyses).

Two further constraints shape any honest solution. First, dense sensing does not scale: monitoring hardware runs to thousands of euros per installed channel, and a 2025 Nature Communications survey found under 20% of even the world's long-span bridges carry any monitoring system; ordinary bridges almost never do. Dresden's Carola Bridge collapsed in 2024 at the one deck not yet refurbished, from hidden prestressing steel corrosion the expert report said visual inspection could not have caught. Partial coverage protects the spans you picked, not the one that fails. Second, the inspection record is itself a weak signal: ratings are coarse (0 to 9), inspector-subjective, and refreshed only every couple of years, so a defect born the day after an inspection waits a long time to be seen.

THE PROBLEM IN ONE SENTENCE

Every one of these collapses generated weak signals in time to act. The gap was never sensing. It was that no system on earth was tasked with noticing that the physical evidence and the official record no longer agreed, and with forcing a decision when they did not.

02 Proposed Solution

DISSENT predicts infrastructure failure by continuously detecting its best documented precursor: the growing disagreement between what an asset's official record claims and what physics quietly shows. For every structure in a national inventory it maintains two independent accounts and alarms, with calibrated statistical confidence, when they diverge.

shared: age, material, design type, span (facts, not opinions)

PAPER WITNESS (what the record claims)

- condition-rating history (NBI / IBMS)
- element data + inspector narratives
- maintenance + construction sign-offs
- quarantined: opinions stay here

PHYSICS WITNESS (what evidence shows)

- InSAR velocity + acceleration per asset
- weather stressors (ERA5), traffic load
- crowdsourced vibration (extension)
- never shown the paper record

DISSENT ENGINE

state dissent + trend dissent + extreme-evidence override

ranked docket: evidence dossier + dated obligation

The dual-witness design. Immutable birth-certificate facts are visible to both witnesses because they are facts, not opinions; everything downstream of human judgement is quarantined inside the Paper Witness, so a disagreement is never an artefact of the witnesses reading the same opinion twice.

How the system runs

#	STAGE	WHAT HAPPENS
1	Ledger	Three decades of public inspection records (623,000+ US bridges in the NBI; 172,517 Indian National Highway structures in IBMS) are parsed into one rating trajectory and event history per asset.
2	Evidence	Zero-hardware physical observables are attached per asset: free Sentinel-1 InSAR time series, ERA5 weather reanalysis, traffic feeds, optional smartphone-trip vibration.
3	Deconfound	Seasonal and weather effects are regressed out of the displacement and vibration series (temperature moves these signals more than early damage does; benchmark datasets exist precisely to validate this step). Only the residuals go forward.
4	Re-inspect	The Blind Re-Inspector predicts each asset's current condition rating from physics alone, with a conformal interval; structurally similar assets pool statistical strength so sparse-data bridges still get a verdict.
5	Score dissent	Three channels: STATE dissent (the record says 6, physics says 3 to 4, with confidence), TREND dissent (a changepoint detector finds an acceleration the record cannot explain), and an extreme-evidence override (one unambiguous signal never waits for agreement). A calibrated fusion yields one dissent score with full evidence attribution.
6	Force action	A fixed alert budget matched to real inspection capacity produces a ranked docket; every flagged asset gets a dossier and a dated obligation, and the field outcome flows back as a training label.

02 / PROPOSED SOLUTION, CONTINUED

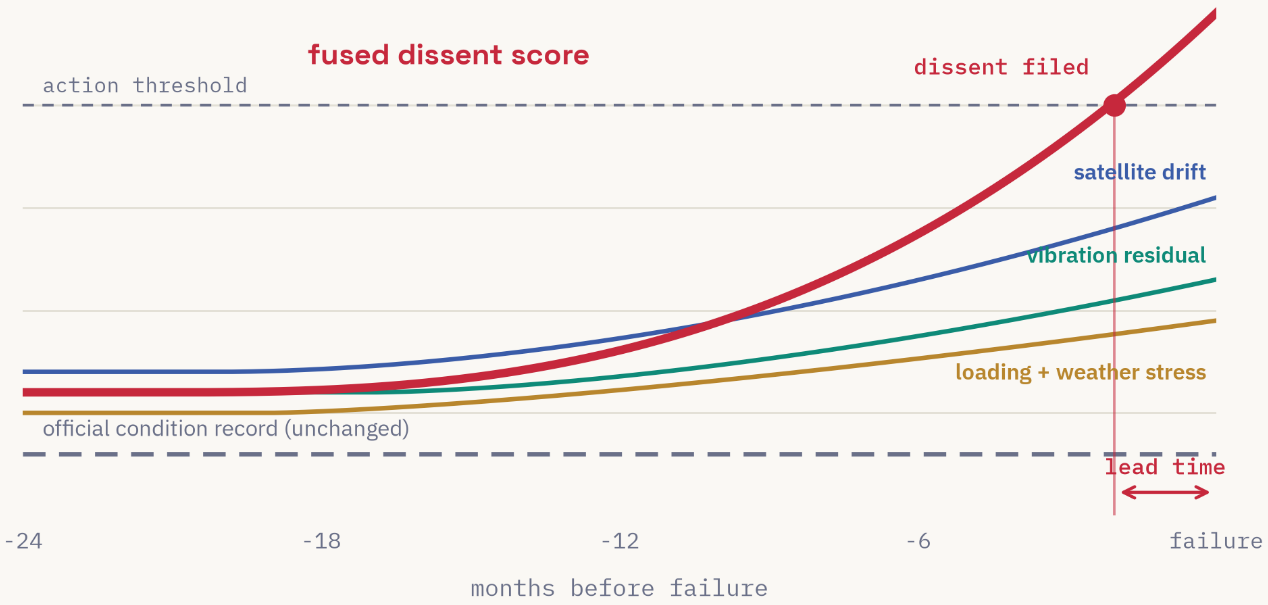


Figure: the signature DISSENT is built to catch, drawn schematically from the published Morandi record. Every individual signal stays below its own alarm threshold and the official record never moves, yet the fused dissent score crosses the action line months before failure. Illustrative schematic, not measured output.

From score to obligation: the action ladder

A ranked list is not a decision, and the forensic record above is a catalogue of correct information that never forced one. Dissent bands therefore carry escalation semantics, and the docket is filled by a small knapsack optimiser that takes the agency's actual monthly inspection capacity as an input, so the system never asks for more field work than exists.

WATCH	Asset chart annotated; dissent trajectory and its evidence attribution visible to the district engineer at next review.
SCHEDULE	A conformal "inspect-by" date is issued, chosen so the statistical coverage guarantee holds; the date, not a score, enters the work plan.
INSPECT NOW	Mandatory targeted inspection within 30 days, scoped to the elements the evidence attribution points at.
RESTRICT	Load-posting or traffic-restriction review is triggered pending an engineer's sign-off; the dossier states exactly which evidence forced the escalation.

The TREND channel and the ladder together also cover the uncomfortable case where paper and physics agree that an asset is bad and it still fails, which is the true story of I-35W: its record said deficient for years. A worsening trend on an already-poor asset climbs the ladder even when state dissent is zero, because the obligation attaches to the trajectory, not only to the contradiction.

03 Uniqueness vs Existing Solutions

Everything on the market today waits for a signal to get loud: a sensor to cross its threshold, a satellite report to show movement, an inspector to write a low rating. DISSENT's output is a different object altogether: a calibrated measurement of disagreement between evidence streams that are individually quiet. To our knowledge no commercial or academic system computes that quantity as its product.

EXISTING APPROACH	WHAT IT DOES	WHERE IT STOPS	DISSENT DIFFERENCE
Dense sensor SHM (Resensys, OSMOS, Worldsensing)	Per-asset strain, vibration and tilt hardware with per-sensor alarm thresholds	Thousands of euros per installed channel; under 20% of even long-span bridges have any SHM; unmonitored spans fail anyway	Zero installed hardware; audits 100% of an inventory from satellite and records
Wide-area InSAR screening (TRE Altamira, SatSense, EGMS)	Millimetre-level displacement maps and periodic reports	Single modality, displacement only, and explicitly no owner: a movement report can land in a folder and stay there	Displacement is cross-examined against the record, weather and traffic, and a dissent forces a dated action
Actuarial risk models (Fracta, NBI deterioration models, AASHTOWare BrM)	Rank asset cohorts from static attributes and biennial ratings	No live sensing, so an individual asset drifting between inspections is invisible	Treats those ratings as one fallible witness and tests them against live physics
Inspection AI (Dynamic Infrastructure, RoadBotics, Niricson)	Computer vision over inspection, drone or vehicle imagery	Cadence bound to inspection campaigns; inherits the inspector's output as ground truth	The inspector's output is the hypothesis under audit; cadence is the satellite revisit, measured in days
Digital-twin platforms (Bentley iTwin, Autodesk Tandem)	Enterprise 3D integration of models and live feeds	The predictive analytics are whatever the owner builds; enterprise pricing excludes small agencies	Ships the analytical core itself, on free open data, runnable by a county engineer
Closest academic analog (Liu and El-Gohary 2022)	Deep fusion of records, traffic, weather and inspection text to forecast condition ratings	Predicts what inspectors will say, faithfully learning their subjectivity and latency	Inverts the target: predicts what inspectors should say from physics, and the residual against what they did say is the product

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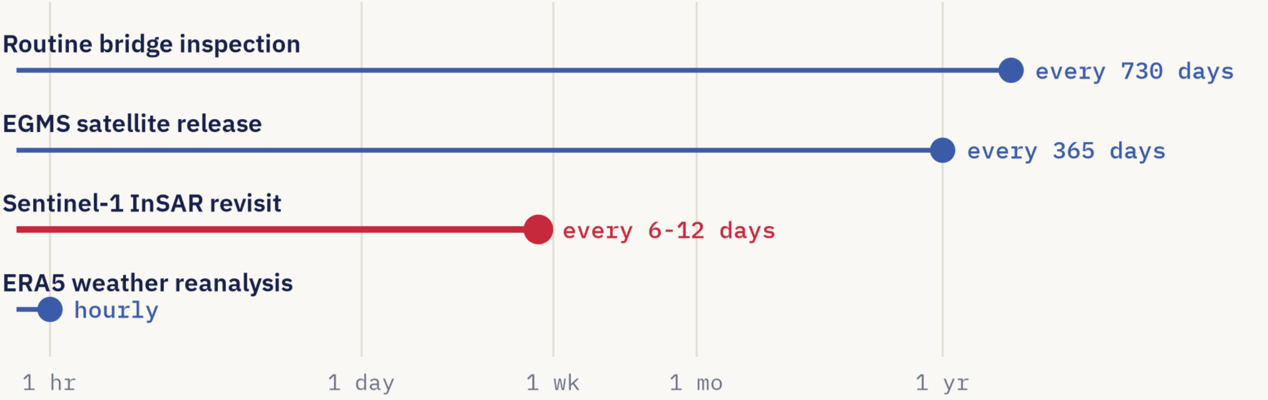
One further difference is statistical, and it is the reason this design is trainable at all. A rival system that tries to predict collapse directly must learn from a few dozen events worldwide per year. DISSENT never trains a rare-event classifier. It trains a condition-rating predictor on millions of (physical evidence, recorded rating) pairs accumulated over three decades of public inspection files, and validates on the thousands of "the record was forced to catch up" events hiding in those same files: bridges that were suddenly closed or downgraded two or more rating steps between inspection cycles. Actual collapses are never trained on; they are held out as replay tests.

04 How the Model Will Be Built

Data sources (all free and downloadable today)

SOURCE	WHAT IT PROVIDES	CADENCE
FHWA National Bridge Inventory + LTBP InfoBridge	Condition ratings 0 to 9 for 600,000+ US bridges across three decades, element-level data, maintenance and reconstruction events; the training labels	Inspection cycle, typically 24 months
Copernicus Sentinel-1 InSAR (EGMS over Europe; LiCSAR products + MintPy processing elsewhere)	Millimetre-scale ground and structure displacement time series per asset footprint, velocity and acceleration	Satellite revisit 6 to 12 days; EGMS releases annually
ERA5 / ERA5-Land reanalysis	Hourly weather since 1940: temperature for deconfounding, freeze-thaw cycle counts, precipitation extremes, soil moisture (flood and scour are the leading US failure cause)	Hourly, about a daily availability lag
Traffic (Caltrans PeMS, TomTom stats, toll and telematics feeds)	Loading proxies and their growth over time	Hourly to monthly
Crowdsourced smartphone vibration (Matarazzo et al. 2022 method)	Bridge modal frequencies recovered from ordinary phone trips with no sensors on the bridge (on the Golden Gate, roughly 80 to 90 trips brought frequency errors to about 3%); planned extension channel	Opportunistic
Z24 and KW51 benchmark datasets	Instrumented bridges with known damage states, used to validate the deconfounding stage, not to train the product	Static benchmarks
India localisation: MoRTH IBMS, Bhuvan/NRSC layers	Structural Rating Numbers and inventory for National Highway structures; the same pipeline re-targets by swapping the ledger	Survey cycles



Days between looks, log scale. The record refreshes in years, satellites in days, weather in hours; nothing in the fusion assumes synchrony, and every channel contributes at its own rhythm.

Build steps

STEP 01	LEDGER BUILD Ingest the historical NBI files into DuckDB with PostGIS geometry; join element data and maintenance events; one trajectory per structure. (Inspector narrative text, where a state DOT publishes it, is embedded with a small language model as an extension channel; the core system does not depend on it.)
STEP 02	PROXY-EVENT MINING Scan the trajectories for record-catches-up events: closures or downgrades of two or more rating steps between cycles, treated as interval-censored (the true change happened somewhere inside the 24-month gap). These become validation positives by the thousands. Sensitivity checks re-run everything under alternative event definitions, since some downgrades reflect policy or coding changes rather than physics.
STEP 03	PHYSICS FEATURES Pull EGMS point time series for a European pilot region and run MintPy small-baseline processing on free LiCSAR Sentinel-1 products for a US corridor subset; aggregate per asset footprint into coherence-weighted robust velocity and acceleration. Attach ERA5 stressor accumulators and traffic loading.
STEP 04	DECONFOUND Per asset, regress displacement and vibration features on temperature and soil moisture harmonics and keep the residuals. Validate on Z24 and KW51 that residual features separate damage states that raw features cannot; this stage is what stops the system rediscovering summer.
STEP 05	BLIND RE-INSPECTOR Gradient-boosted model from physics-only features to current condition rating, wrapped in split-conformal intervals. Evaluation uses rolling-origin temporal splits plus asset-level holdouts, so repeated inspections of one structure never straddle train and test, and a population-level mapping is what an individual asset is judged against.
STEP 06	DISSENT CHANNELS STATE dissent: recorded rating minus the upper conformal bound of the physics-predicted rating, required to persist across cycles and exceed the known inspector-variability noise band before it counts. TREND dissent: Bayesian online changepoint detection on the deconfounded residuals, tuned for acceleration onsets. OVERRIDE: an unambiguous extreme signal bypasses fusion entirely. Isotonic-calibrated fusion produces the final score with per-feature evidence attribution.
STEP 07	REPLAY VALIDATION Freeze the pipeline and replay the published pre-collapse records: Morandi (under both the Milillo and the contesting Lanari processing analyses, reporting whether a dissent fires under each) and Surfside (published subsidence study against the deferred-repair paper trail), plus I-35W as a records-and-trend-only case with no usable satellite channel.
STEP 08	DOCKET PRODUCT A lightweight triage interface: ranked docket, per-asset chart showing the dissent trajectory and which evidence moved it, auto-generated dossiers, and the write-back path that turns every field verification into a label.

Honest engineering constraints

Three risks are named up front rather than discovered by a reviewer. First, InSAR coverage: many bridges present few coherent radar scatterers (vegetation, small decks, unfavourable geometry), so every asset carries per-modality quality flags and the models are trained with modality dropout; an asset with no usable satellite channel degrades gracefully to records, weather and traffic scoring instead of silently vanishing. Second, the STATE channel's power depends on how well physics-only features predict a coarse, noisy 0-to-9 label; published deterioration models built on records alone typically top out near a 0.70 concordance index, so conformal intervals may be wide, and the design explicitly lets TREND dissent carry the load where STATE dissent is uninformative. Third, cadence: the record refreshes in years, satellites in days, weather in hours. Nothing in the fusion assumes synchrony; every channel contributes at its own rhythm.

Evaluation plan

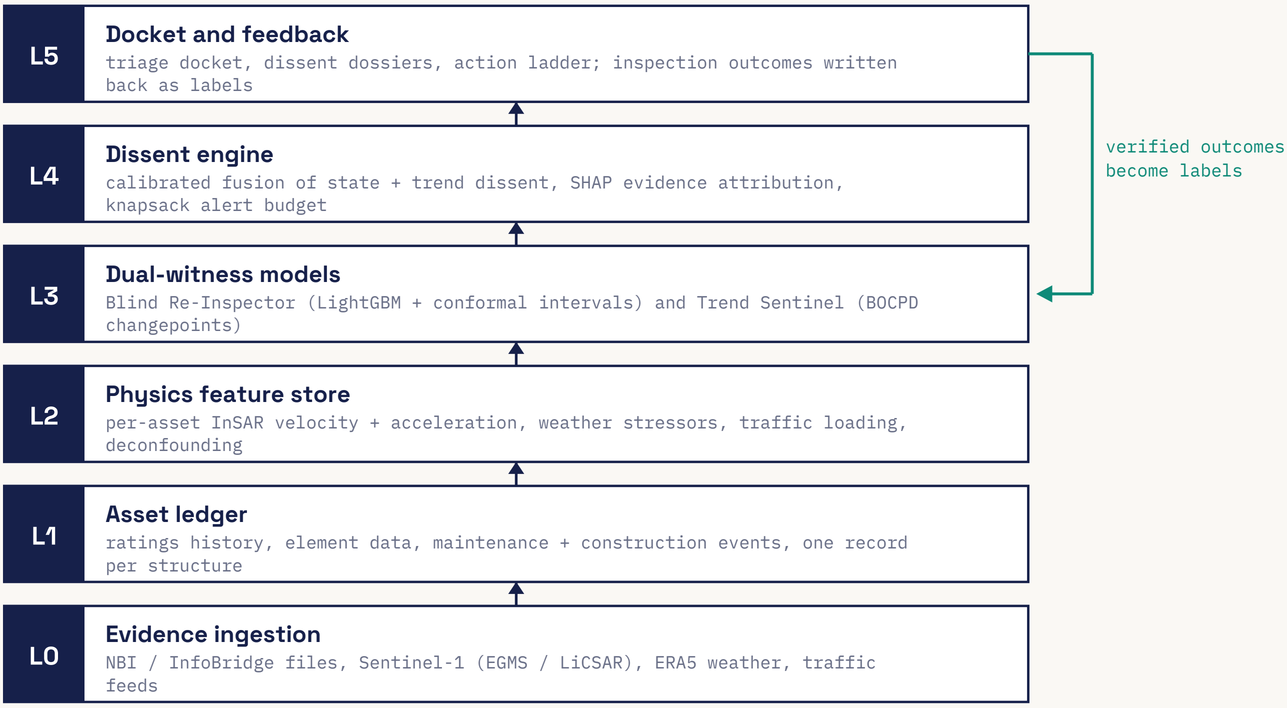
METRIC	WHAT IT ESTABLISHES
Event recall and lead-time distribution on held-out proxy events	The headline: what fraction of real sudden-downgrade events the system flagged early, and by how many months. Target for the pilot: flag at least 60% of two-step sudden downgrades at least 6 months early within the alert budget.
PR-AUC of fused score vs two ablations (physics-only anomaly detector; paper-only deterioration model), plus STATE-only vs TREND-only vs fused	Proves the disagreement itself carries the signal, which is the whole uniqueness claim; precision-recall rather than ROC because events are rare.
Recall at fixed alert budget; false alerts per thousand asset-months	Shows the docket is absorbable by a real inspection team, not an alarm firehose.
Conformal coverage and calibration (reliability diagrams, Brier score)	Makes each dissent decision-grade: the quantitative forcing threshold the FIU morning meeting never had.
Named-case replays (Morandi under both processing chains, Surfside, I-35W)	Judge-legible validation on the most famous collapses with published pre-failure data.
Deconfounding efficacy on Z24/KW51; concordance of time-to-downgrade ranking	Pre-empts the standard expert objections: temperature confounding and actuarial-baseline comparison (does the dissent signal add skill over the roughly 0.70 static-covariate ceiling?).

The pilot commits to numbers up front, so the evaluation section pays off a promise made on page one:

60%	**6 mo**	**0.5%**	**3**
2-STEP SUDDEN DOWNGRADES FLAGGED EARLY (TARGET)	MINIMUM EARLY-WARNING LEAD TIME (TARGET)	ALERT BUDGET PER QUARTER, CAPPED TO REAL CAPACITY	NAMED-CASE REPLAYS: MORANDI, SURFSIDE, I-35W

05 Proposed Architecture

Six layers, all open source, sized so that a one-region pilot runs on a single laptop with no cloud spend. Evidence flows up; verified outcomes flow back down as labels.



DISSENT layered architecture. The teal return path is the loop the forensic record says was always missing: field verification feeding back into the model.

L0, ingestion: scheduled pullers for NBI and InfoBridge files, EGMS or LiCSAR Sentinel-1 granules, ERA5 via the Climate Data Store API, and traffic feeds, orchestrated with Prefect and aligned to the satellite revisit. L1, asset ledger: DuckDB and PostGIS, keyed by structure number (NBI) or IBMS identity in the India build: the Paper Witness lives here. L2, physics store: per-asset displacement velocity and acceleration, weather stressors, loading, then the deconfounding module: the Physics Witness's evidence. L3, models: the Blind Re-Inspector (LightGBM plus MAPIE conformal wrappers, with population-based transfer so structurally similar bridges pool strength) and the Trend Sentinel (Bayesian online changepoint detection). L4, dissent engine: calibrated fusion, evidence attribution, the alert-budget governor and the action ladder. L5, docket: the triage interface, dossier generator and outcome write-back.

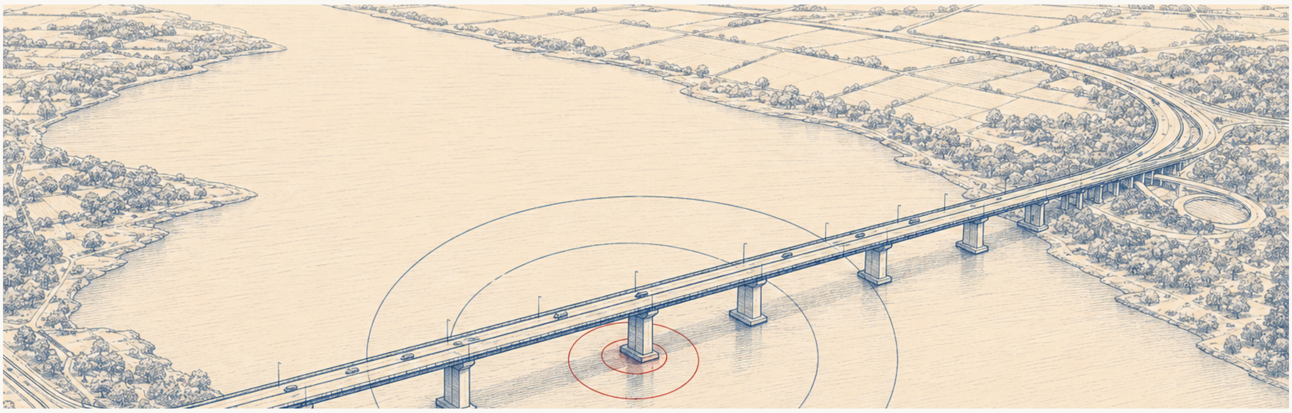
Staged build (what we would actually do first)

PHASE	SCOPE	EXIT TEST
v1 one semester	One pilot inventory (a single US state via LiCSAR corridor subset, or a European region via EGMS); records + satellite + weather channels; both dissent channels; Morandi and Surfside replays; docket demo	Replays fire with stated lead time; proxy-event recall beats both ablation baselines
v2 scale-up	Full-state satellite processing, traffic feeds, crowdsourced vibration pilot, text channel where a DOT publishes narratives, India IBMS localisation	Recall at fixed alert budget holds at inventory scale; a partner district consumes the docket in its real work plan

05 / STAGED BUILD : THE V1 SEMESTER, WEEK BY WEEK

WEEKS	V1 MILESTONE
1 to 3	Ledger built; actuarial baselines reproduced (Kaplan-Meier, Cox, gradient boosting on static covariates) so the dissent signal has an honest floor to beat
4 to 6	Satellite channel extracted for the pilot inventory; coverage and quality flags measured and reported
7 to 8	Deconfounding validated on Z24 and KW51
9 to 10	Blind Re-Inspector trained with conformal calibration; dissent channels fused
11 to 12	Replay gauntlet run; docket interface demo; pilot report written

06 Real-World Impact



One satellite pass, every asset on the corridor audited. Illustration generated for this report.

The United States inspects most bridges on a 24-month cycle, so a defect born the day after an inspection waits, on average, a year to be seen. DISSENT compresses that discovery latency toward the satellite revisit, measured in days, for every structure in an inventory at once, at a marginal cost near zero. The scale is documented: the 2025 ASCE Report Card grades US bridges C, with 6.8% of 623,218 bridges in poor condition and a projected 373 billion dollar ten-year investment need, while ARTBA's 2025 analysis counts 163 million daily crossings over the 41,600 poor-rated bridges. No inspection workforce can look everywhere; DISSENT tells it where physics is quietly disagreeing with the paperwork, which is the highest-value place to look next.

730 d TODAY: AVERAGE GAP BETWEEN LOOKS	**6-12 d** WITH DISSENT: SATELLITE REVISIT CADENCE	**100%** OF AN INVENTORY AUDITED, ZERO INSTALLED HARDWARE	**~$0** MARGINAL DATA COST: ALL SOURCES ARE OPEN

It is designed to be adopted, not resented. The system is a second opinion that protects inspectors and re-aims their scarce hours: their findings are the very labels that train it, their verifications improve it, and the dossier language is evidence-first rather than blame-first. For small agencies the equity effect is real: a county with five bridges and one engineer gets the same quality of second opinion as a national railway, because every input is free and the pilot stack runs on a laptop.

The India build is not an afterthought but a deliberate second target, and the timing is exact. IBMS already inventories 172,517 National Highway structures with condition ratings, and its first survey cycle identified 147 critically distressed bridges. In July 2026 MoRTH launched a mandatory nationwide digital re-survey of all of them through a new IBMS mobile app, with a September 2026 completion deadline: that survey is running right now, and it is creating precisely the fresh, centralised paper baseline that DISSENT is built to audit against physics. Morbi remains the canonical case of the failure mode in question, a renovation record that nobody checked against physical reality. The pipeline localises by swapping the ledger source and adding Bhuvan and NRSC satellite layers; nothing in the method is US-specific.

What DISSENT cannot do, stated plainly

A failure with no precursor in any open channel is invisible to it: sudden scour during a flood peak between satellite passes, or construction-phase errors on structures too young to have a record, as at FIU. Assets with no usable radar coverage fall back to a weaker records-and-weather audit. The Morandi satellite precursor itself is scientifically contested, which is precisely why the replay is run under both published analyses and why no single channel is load-bearing. We consider naming these limits a feature of the report, not a weakness of the system: the field's history is full of confident tools, and DISSENT's entire premise is scepticism about confident records.

The long-term payoff compounds. Every dissent verified in the field becomes a labelled precursor, and after a few years of operation the system will have assembled something that does not currently exist anywhere: a prospective, multi-modal dataset of what infrastructure failure looks like before it happens. That dataset, more than any single alarm, is what moves the whole field from forensics to foresight.

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TEAM NEXUS NETWORK

AI INNOVATION CHALLENGE 2026 | AI FORGE ROUND 2 | 25 AUGUST 2026